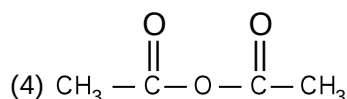
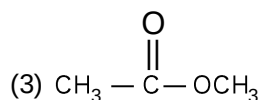
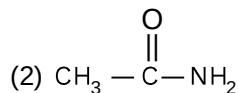
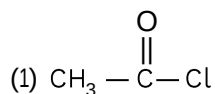
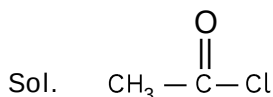


CHEMISTRY

1. Which of following molecules hydrolyse fastest?



Ans. (1)



2. Which of the following statements is/are correct for the adiabatic process?

(a) Molar heat capacity is zero

(b) Molar heat capacity is infinite

(c) Work done on gas is equal to increase in internal energy

(d) The increase in temperature results in decrease in internal energy

(1) (a) and (c) only

(2) (b) and (c) only

(3) (a) and (d) only

(4) (c) and (d) only

Ans. (1)

Sol. $C_m = \frac{q}{n\Delta T}$

For adiabatic process $q = 0$ $C_m = 0$

$$\Delta U = q + w$$

$$\Delta U = w$$

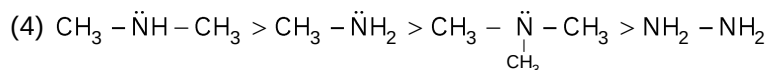
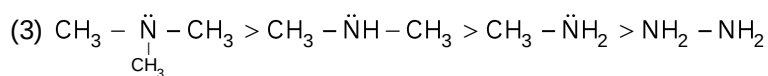
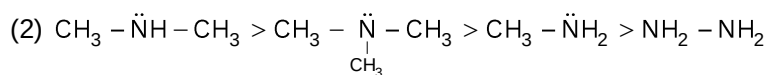
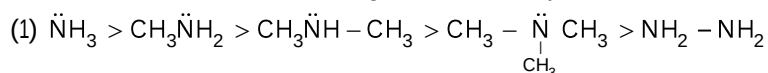
Work done on gas = +ve

$$\Delta U = +ve$$

$$T \uparrow \quad \Delta U \uparrow$$

So option (1) Correct.

3. Which is Correct basic strength order in aqueous medium



Ans. (2)

Sol. Basic strength order for $R = CH_3$
 $2^\circ \text{ amine} > 1^\circ \text{ amine} > 1^\circ \text{ amine} > NH_3$
 so option (2) correct

4. In AX_4Y , X and Y are the top two electronegative atoms and A has the highest ionization enthalpy in its period.

Predict the hybridization of AX_4Y .

- (1) sp^3
- (2) sp^3d
- (3) sp^3d^2
- (4) sp^2

Ans. (3)

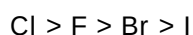
Sol. A has highest ionization in its period so it is a Noble gas and X & Y top electronegative atoms fluorine and oxygen. So the molecule may be $XeOF_4$ having sp^3d^2 hybridisation.

5. Which of the following property does not have the regular trend in group 17th of period table.

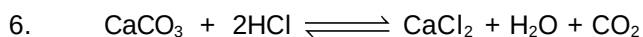
- (1) Atomic size
- (2) I.E.
- (3) Electronegativity
- (4) Electron Affinity

Ans. (4)

Sol. Electron affinity has the trend



(Irregular trend)



1000g 2 l, 2M

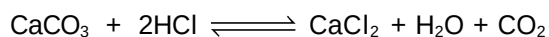
Find moles of $CaCl_2$ reduced.

- (1) 10
- (2) 2
- (3) 6
- (4) 8

Ans. (2)

Sol. Moles of $CaCO_3 = \frac{1000}{100} = 10$

Moles of $HCl = 2 \times 2 = 4$



$$10 \quad 4 \text{ (LR)} \quad \frac{1}{2} \times 4$$

$$= 2 \text{ moles}$$

2 moles, correction option : (2)

7. Statement-I: Ga has larger metallic radius than Al.

Statement-II: Ga^{+3} has larger ionic radius than Al^{+3} .

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are false

(2) Statement-I is false but Statement-II is true

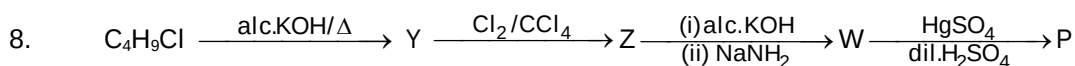
(3) Both Statement I and Statement-II are true

(4) Statement-I is true but Statement-II is false

Ans. (2)

Sol. Ga (135 pm) Al (143 pm)

(Size) $\rightarrow \text{Ga}^{+3}$ (62pm) Al^{+3} (53.5 pm)



What is the correct IUPAC name of major product 'P'

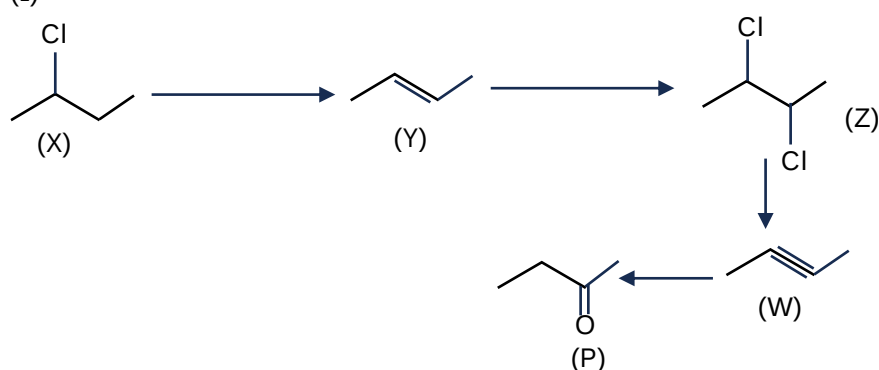
(1) Butan-2-one

(2) Butan-2-ol

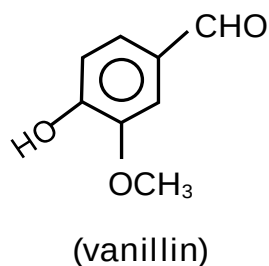
(3) n-butane

(4) Butanal

Ans. (1)

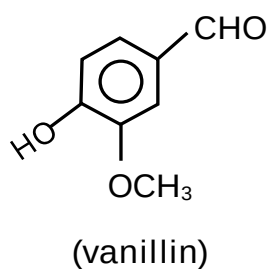


Sol.



9. Statement-I:

Gives self aldol condensation.



Statement-II:

Gives Positive tollen's test.

- (1) Both Statement – I & II are correct
 (2) Both Statement – I & II are Incorrect
 (3) Statement – I is correct & Statement II is Incorrect
 (4) Statement – I is Incorrect & Statement II is correct

Ans. (4)

Sol. For self-aldol α -H on sp^3 carbon should be present.
 For Tollen's test aldehyde, (aliphatic or aromatic) is present
 So in statement 2, is aromatic aldehyde hence correct.
 And statement-1 is wrong.
 Correct option 4.

10. Which is correct option about radius of n^{th} orbit in H-atom.

- (1) Radius of 3rd orbit is 9 times of radius of 1st orbit
 (2) Radius of 4th orbit is 4 times of radius of 1st orbit
 (3) Radius of 8th orbit is 4 times of radius of 2nd orbit
 (4) Radius of 6th orbit is 2 times of radius of 3rd orbit

Ans. (1)

Sol. Radius of n^{th} orbit (r_n) = $a_0 \frac{n^2}{Z}$

For H-Atom, $r_n = a_0 n^2$

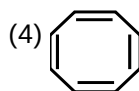
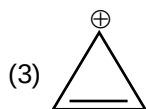
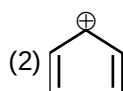
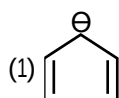
Option 1 : $r_3 = a_0(3)^2 = 9a_0$

$r_1 = a_0(1)^2 = a_0$

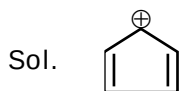
So $r_3 = 9r_1$

Correct option : (1)

11. Out of the following species, which one is anti-aromatic?



Ans. (2)



$$4n = 4$$

$$n = 1$$

so antiaromatic

correct answer option (2)

12. Which of the following option is correct?

- (1) Cysteine (amino acid) react due to its free sulphur atom.
- (2) Except glycine, all α -amino acids have only one chiral center.
- (3) All α -amino acids are optically active.
- (4) Only glutamic acid having —COOH group in parent chain.

Ans. (1)

Sol. Fact

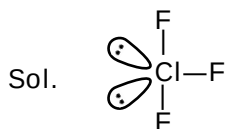
13. Among the given compounds



Find the hybridization of the compound which is polar & having highest number of lone-pairs on central atom.

- (1) sp^3d
- (2) sp^3d^2
- (3) sp^3d^3
- (4) sp^3

Ans. (1)



14. Two volatile liquids A & B Form ideal solution. 1 mol of A is mixed with 3 moles of B. Total vapour pressure was 500 torr.

If vapour pressure of A was 200 torr. Select correct option.

- (1) B is more volatile & vapour pressure of B is 600 torr.
- (2) B is more volatile & vapour pressure of B is 450 torr.
- (3) A is more volatile & vapour pressure of B is 600 torr.
- (4) A is more volatile & vapour pressure of B is 450 torr.

Ans. (1)

Sol. $P_T = X_A P_A^0 + P_B^0 X_B$

$$500 = \frac{1}{4} \times 200 + \frac{3x}{4}$$

$$500 = 50 + \frac{3x}{4}$$

$$4 \times 450 = 3x$$

$$x = \frac{4 \times 450}{3} = 600 \text{ torr}$$

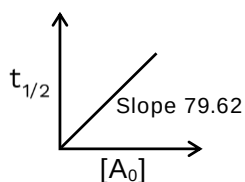
15. Which of the following is incorrect order of energy?

- (1) GS. of H and 1st ES of He^+
- (2) GS. of H and 2nd ES of Li^{2+}
- (3) 1st ES of He^+ and 2nd ES of Li^{2+}
- (4) GS. of He^+ and 1st ES of Li^{2+}

Ans. (4)

Sol. $E \propto \frac{Z^2}{n^2}$

16. In the following graph between $t_{1/2}$ and initial concentration $[A_0]$. If slope of the graph is $79.62 \text{ M}^{-1} \text{ min.}$ and initial concentration is 2.5 M . Find the concentration of A after 10 min.



Ans. (2)

Sol. For zero reaction, $t_{1/2} = \frac{A_0}{2K}$; $K = \frac{1}{2 \times 79.62}$

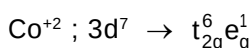
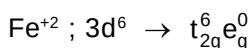
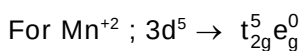
$$\therefore C_0 - C_t = Kt$$

$$\text{or, } C_t \approx 2$$

17. Metal having maximum standard reduction potential among Cr, Mn, Fe, Co form an octahedral complex $[M(\text{CN})_6]^{-4}$. Number of electrons in e_g orbital is

Ans. (1)

Sol. CN^- is a SFL



18. $\text{H}-\overset{\text{a}}{\text{C}}=\overset{\text{b}}{\text{C}}-\overset{\text{c}}{\text{CH}_2}-\overset{\text{d}}{\underset{\text{H}}{\underset{\text{CH}_3}}{\text{CH}}}-\text{CH}_2$ Which form most stable & least stable free radical respectively.

(1) b,a

(2) c,d

(3) a,c

(4) a,d

Ans. (1)

Sol. (a) free radical in sp^2 carbon.

(b) Resonance stabilised